

The effects of cations and anions on the ionic conductivity of poly[bis(2-(2-methoxyethoxy)ethoxy)phosphazene] doped with lithium and magnesium salts of trifluoromethanesulfonate and bis(trifluoromethanesulfonyl)imide

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ABSTRACT

Lithium trifluoromethanesulfonate (LiTf), lithium bis(trifluoromethanesulfonyl)imide (LiTFSI), magnesium trifluoromethanesulfonate (MgTf₂) and magnesium bis(trifluoromethane sulfonyl)imide (MgTFSI₂) were dissolved in poly[bis(2-(2-methoxyethoxy)ethoxy)phosphazene] (MEEP) to compare the effect on solvent-free polymer ionic conductivity of monovalent versus divalent cations, and two anions with different degrees of dissociation. The polymer electrolytes with the bis(trifluoromethanesulfonyl)imide anion had higher ionic conductivities even though the glass transition temperatures, which reflected polymer molecular motion, were higher than those of their counterparts with the trifluoromethanesulfonate anion. Furthermore, polymer electrolytes with magnesium salts achieved their maximum conductivity at lower salt concentrations than their counterparts with lithium salts. The temperature dependence of the ionic conductivity of the solid solutions was fitted to the Vogel–Tamman–Fulcher (VTF) equation. The pseudo-activation energy term, *B*, of the VTF equation showed a strong dependence on the anion present. This result suggests that the dominant mobile species is the anion, while the cation remains relatively bound to the polymer.

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1. Introduction

The mechanism of ion transport in solid and gel polymer systems is a subject of widening interest since lithium-type rechargeable batteries have assumed a prominent role in electronic devices and future automotive technology. This subject has been the focus of considerable debate that started with the original discovery by Wright [1] of ion transport in poly(ethylene oxide) (PEO), and the subsequent recognition by Armand [2] that polymeric ionic conductors have the potential to replace or supplement the existing technology based on organic solvent-based electrolytes. The advantages of polymer electrolytes over liquid-based systems include resistance to combustion, robustness under challenging conditions, and access to compact energy storage devices that can be fabricated into intricate shapes [3].

The main disadvantage of polymer electrolytes is their low ionic conductivity when compared to liquid electrolytes. This is due to the slower dynamics of polymer chain motion compared to small molecules in the liquid state. An unplasticized PEO system has an ambient temperature conductivity of 10^{-6} S/cm mainly due to high

levels of crystallinity. This conductivity does not meet the minimum requirement of 10^{-3} S/cm for practical energy storage applications. Poly[bis(2-(2-methoxyethoxy)ethoxy)phosphazene] (MEEP) (Fig. 1) is an alternative, totally amorphous polymer that has been studied intensively as a polymer electrolyte since its initial synthesis and evaluation in the laboratories of Allcock and Shriver in 1984 [4–6]. Studies have shown that room temperature ionic conductivities as high as 10^{-4} S/cm are possible even without plasticizers. This is several orders of magnitude higher than the value for PEO, but is still lower than the optimum values. The higher conductivity of MEEP electrolytes is attributed to the flexible phosphorus–nitrogen backbone and the oligomeric etheric side chains, both of which are responsible for the low glass transition temperature (*T_g*) of -80 °C. Although the low *T_g* gives MEEP poor dimensional stability, cross-linking of this polymer yields a dimensionally stable material without a significant sacrifice of ionic conductivity [7].

The conductivity of lithium salts in SPEs including MEEP has been studied extensively because lithium has the highest electrode potential, which leads to the prospect of high energy density batteries [8]. However, lithium metal is reactive if the cell is breached. In the search for alternatives, metals with multivalent cations have been considered. The rationale for studying the ionic conductivity of multivalent cations is that they might generate higher conductivities

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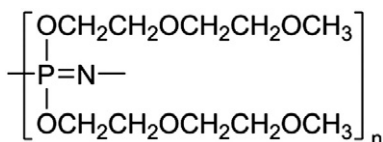


Fig. 1. Poly[bis(2-(2-methoxyethoxy)ethoxy)phosphazene] (MEEP).

than monovalent cations since each cation carries more than one charge [9]. Among the possible multivalent cations, Mg^{2+} was chosen for this study due to the similar chemical hardness and size compared to Li^+ . Therefore, in principle the cation–polymer interactions should be similar for both, although the charges are different. The conductivity of Mg^{2+} in polymer electrolytes has been studied previously with other polymer systems and its use in battery applications has been considered [10,11]. Although magnesium does not have as high an electrode potential as lithium, the cost and safety of using magnesium could outweigh the electrode potential disadvantage.

Salts of trifluoromethanesulfonate (Tf) are commonly used in polymer electrolyte studies because the negative charge on the Tf ion is delocalized over the sulfonate group. Hence, the salt dissociates easily. However, bistrifluoromethanesulfonimide (TFSI), shown in Fig. 2, is a “new generation” anion that has more extensive charge delocalization than Tf, and it promotes a higher degree of dissociation [12].

One of the main questions about the mechanism of ionic conduction in polymers such as PEO or MEEP is the specific role played by the anions and cations. Most polymer electrolyte studies report the bulk ionic conductivity. However, ionic conduction in a polymer is the result of contributions from both cations and anions because both ions carry charge and are mobile [13]. Some investigators have favored the view that the cations carry most of the charge, while others have identified the anions as the main charge carriers. Therefore, it is important to differentiate their separate contributions, especially when evaluating the performance of a polymer electrolyte for a rechargeable lithium battery [14]. Different types of measurements, different salts, or different polymers often favor one mechanism or the other. Furthermore, the cationic and anionic conduction mechanisms are not the same. The study reported here was designed to throw further light on this situation, especially with respect to the roles played by Tf and TFSI anions and the consequences of replacing lithium cations by magnesium. Thus, we have examined the effect of two different cations and two anions on the ionic conductivity of MEEP and have identified the dominant mobile species.

2. Experimental

2.1. Materials

All materials were used as received unless stated otherwise. Sodium (ACS reagent grade), lithium trifluoromethanesulfonate (99.995%), magnesium trifluoromethanesulfonate (97%), lithium bis(trifluoromethanesulfonyl)imide (99.95%), bis(trifluoromethanesulfonyl)imide (95%), and magnesium acetate tetrahydrate (98%) were obtained from Aldrich and were stored in an argon atmosphere glove box. 2-(2-Methoxyethoxy)ethanol (99.0%) was obtained from Aldrich and was purified by drying over CaH_2 and distillation under reduced pressure. Hexachlorocyclotriphosphazene was obtained from Fushimi Pharmaceutical Co. Ltd. (Japan) and was purified by recrystallization twice from heptanes followed by sublimation at 30 °C at 0.01 mm Hg. Tetrahydrofuran (99.99%, EMD) was dried by passage through Glass Contour alumina columns before use. De-ionized water was obtained from a Barnstead Nanopure Diamond water purification system. All synthesis reactions were carried out under an atmosphere of dry nitrogen.

2.2. Equipment

High-field ^1H (360.14 MHz) and ^{31}P (145.79 MHz) NMR spectra were obtained with the use of a Bruker AMX-360 NMR spectrometer. ^1H NMR spectra were referenced to internal tetramethylsilane, while ^{31}P NMR spectra were proton-decoupled and referenced to external 85% phosphoric acid in D_2O . Mass spectra were collected using a Micromass Quattro-II triple quadrupole mass spectrometer. Polymer molecular weights and polydispersities were estimated using a Hewlett-Packard HP 1090 gel permeation chromatograph equipped with an HP-1047A refractive index detector, Phenomenex Phenogel 10 μm linear columns were calibrated against polystyrene standards. Sample elution was carried out at 40 °C with a 0.1 wt.% solution of tetra-*n*-butylammonium nitrate in THF. Thermal transitions were identified with use of a TA Q10 differential scanning calorimeter. Calibration was accomplished with indium, water, and cyclohexane standards. All analyses were performed over a range of –100 to 100 °C at a heating rate of 10 °C/min. Ionic conductivity measurements were obtained using a Hewlett-Packard 4192A LF impedance analyzer at a potential of 0.1 V with an alternating current frequency range of 5 Hz–1 MHz. The samples were placed between platinum electrodes with a Teflon O-ring spacer, and the polymer electrolyte cell was compressed between aluminum blocks held in a Teflon fixture. Electrical leads were attached between the impedance

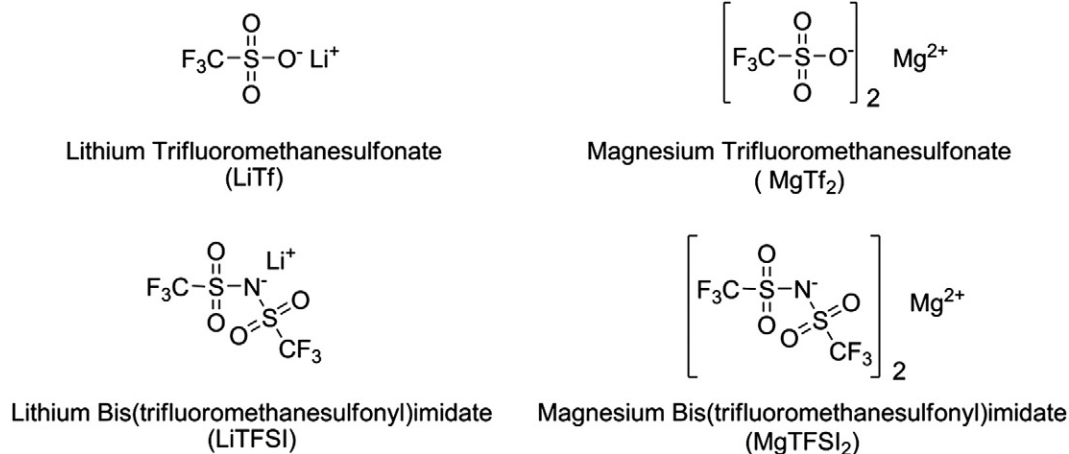


Fig. 2. Salts used in this study.

analyzer and the polymer electrolyte cell sample holder. All ionic conductivity measurements were carried out over a temperature range of 25–80 °C under an inert atmosphere of dry argon.

3. Syntheses

Poly[bis(2-(2-methoxyethoxy)ethoxy)phosphazene] (MEEP) was synthesized using previously published procedures [4,5].

3.1. Magnesium bis(trifluoromethanesulfonyl)imide

Magnesium acetate tetrahydrate was allowed to react with 2 equivalents of bis(trifluoromethanesulfonyl)imide in water under ambient conditions for an hour [15]. The water was removed by rotary evaporation. The MgTfSI_2 was dried further in vacuum for 5 days at 40 °C. The product composition was confirmed by ^{19}F NMR and mass spectrometry.

3.2. Polymer electrolyte fabrication

The salts were added to MEEP dissolved in stirred THF until a homogenous, clear solution was obtained. The THF was allowed to evaporate and residual solvent was removed by drying under vacuum at 40 °C for 5 days.

4. Results and discussion

4.1. Glass transition behavior

The conduction of ions in a polymer electrolyte depends partly on the flexibility of the polymer chains. Flexible macromolecules allow faster ion migration than rigid polymers, and consequently give higher ionic conductivities. The flexibility of the polymer chain is related to the glass transition temperature (T_g), the point at which the polymer undergoes a phase transition from a glassy solid to a rubbery material. The relationship between the T_g of MEEP and the concentration of different salts is shown in Fig. 3. The x-axis of the graph represents the salt concentration in the form of the mole fraction of salt with respect to the repeating units of MEEP. The y-axis of the graph gives the T_g of MEEP in degrees Celsius. The addition of salt to MEEP increased the T_g in a roughly linear fashion with respect to the concentration of the salt. The T_g s of SPEs that contained Mg salts were higher than those with Li salts. MEEP with TFSI salts also had higher T_g s compared to Tf salts with the same cation.

For the formation of a solid polymer electrolyte, the polymer must function as a solvent and favor dissociation of the salts. The dissolution of salts in the polymer matrix produces cations, anions, and also anion–cation associated species. The ability of a polymer elec-

trolyte to dissolve salts depends on the ability of the polymer to stabilize ions by solvation. In MEEP, the cations are stabilized by coordination to the oxygen atoms in the side chains and possibly to the nitrogen atoms in the backbone [16]. Interactions between the cation and the polymer can be understood according to the Hard/Soft acid base (HSAB) principle, which dictates that hard non-polarizable acids interact strongly with hard non-polarizable bases. Mg^{2+} and Li^+ are hard cations with similar hardness values of 35.12 η and 32.55 η respectively [17]. Hence, Mg^{2+} and Li^+ would interact with MEEP similarly, with a stronger interaction to the hard oxygen of the side chains than to the soft nitrogen atoms of the backbone. However, the divalent cation Mg^{2+} has a higher coordination ability compared to the monovalent Li^+ cation. Therefore, MEEP with dissolved Mg^{2+} salts has higher T_g s than with comparable concentrations of Li^+ salts. Cation coordination to the side chain oxygen atoms forms inter and intrachain transient crosslinks. However, studies by Stewart [18] and ourselves have shown that crosslinking via the side chains does not necessarily contribute to an increase in T_g . In our studies, covalent crosslinks between the side chains induced by γ irradiation of MEEP did not raise the T_g significantly even when 1 crosslink was formed for every 1.4 polymer repeat units [19]. The reason is because the etheric side chains are sufficiently long and flexible that crosslinks do not impede the flexibility of the backbone. This implies that any T_g increases in the presence of a dissolved salt are connected with cation coordination to the inner oxygen atoms or to the skeletal nitrogen atoms which reduce the flexibility of the backbone.

Unlike cations, anions are not expected to coordinate to the polymer electrolyte. Nevertheless, our results show that there is a difference in T_g when different anions are used. It has been proposed that TFSI has a plasticization effect on polymers such as PEO by reducing both the crystallinity and the T_g [20]. The same effect was not detected with MEEP because pure MEEP is totally amorphous. In fact, the T_g s of MEEP SPEs that contain TFSI salts were higher than those that contained Tf salts. These higher T_g s can be explained by a greater degree of dissociation of TFSI salts because of their lower lattice energies than Tf salts. Frech and coworkers used diglyme and LiTfSI as a model system to study ion dissociation in etheric polymers, and showed that LiTfSI is fully dissociated when the O:Li ratio is more than 6:1 [21]. In a separate study by the same group using diglyme and LiTf , only 23% of the salt was fully dissociated even when the O:Li ratio was 40:1 [22]. The remainder are associated ionic species such as ion pairs $[\text{LiTf}]$ and triple ions species $[\text{Li}_2\text{Tf}]^+ / [\text{LiTf}_2]^-$. Therefore, the higher degree of dissociation of TFSI salts generates more cations that coordinate to the polymer. This outweighs the plasticizing effect that was detected in PEO, and results in higher T_g s for MEEP with TFSI salts.

4.2. Ionic conductivity behavior

Figs. 4–7 show the ionic conductivities of MEEP with each salt at temperatures from 25 to 80 °C. The x-axis of each graph represents the salt concentration in molar fraction, X_{salt} , and the y-axis shows the ionic conductivity in terms of $\log \text{ S/cm}$. The ionic conductivity of MEEP increased initially with increasing salt concentration until a maximum was reached. Further increases in salt concentration lowered the ionic conductivity. Table 1 shows the maximum room temperature conductivity of MEEP with each salt. Salts with the same anion showed similar maximum conductivities. Even though the MEEP systems containing the TFSI salts had higher T_g s, nevertheless the ionic conductivities were higher than for MEEP electrolytes with Tf salts.

We favor the following interpretation for the initial increase and subsequent decrease of ion conductivity as a function of salt concentration. The ionic conduction increases initially because the number of charge carriers rises with salt concentration. However, increasing salt concentrations also increase the T_g . This lowers the ionic conductivity because a higher T_g corresponds to lower polymer

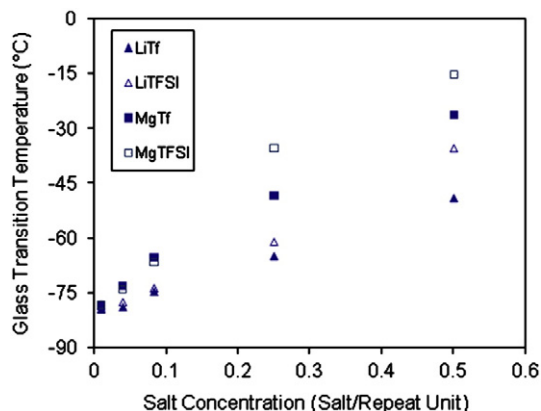


Fig. 3. Glass transition behavior of MEEP with increasing salt concentration.

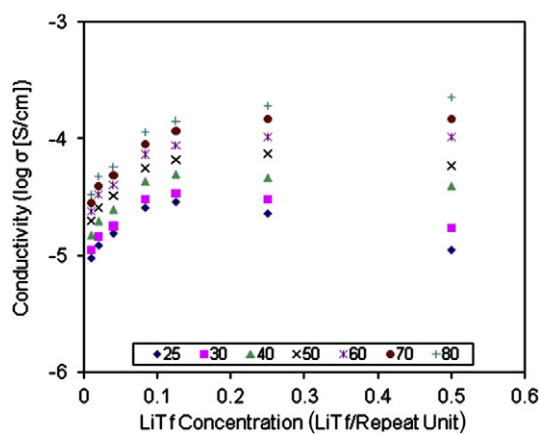


Fig. 4. Isothermal ionic conductivity of LiTf:MEEP.

chain mobility. The decrease in ionic conductivity can also be attributed to the formation of contact ion pairs and triple ion species. As the salt concentration increases, the average distance between ionic species also decreases, which leads to the formation of associated ionic species as mentioned before. Frech and coworkers carried out a study of LiTf dissolved in MEEP and showed that at a MEEP:Li ratio of 8:1, only 21% of the LiTf was dissociated and the remainder consisted of aggregated species [23]. It was proposed that associated ionic species are likely to be immobile because [LiTf] is neutral and does not experience any force under an electric field whereas $[\text{LiTf}_2]^-$ or $[\text{Li}_2\text{Tf}]^+$ may have limited mobility due to its mass. It was postulated that these associated species can still contribute to ionic conduction by functioning as hopping sites for ions, albeit at a slower rate. It is believed that the low conductivity of polymer electrolytes with MgTf_2 salt is due to the extensive formation of associated species even at low concentrations. Hence, the ionic conductivity of MEEP containing MgTf_2 starts to decrease at a much lower concentration compared to the other three systems. No appreciable conductivity was detected at a MEEP: MgTf_2 ratio of 2:1.

As shown in Table 1, the maximum conductivities of polymer electrolytes with the same anion were essentially the same. Furthermore, the polymer systems that contain TFSI salts have ionic conductivities that are almost one order of magnitude higher than their counterparts with Tf salts. Although other polymers with higher T_g s often give lower conductivities, the opposite was found in the MEEP system where polymer solutions of TFSI salts had higher T_g s and also higher conductivities. This phenomenon is probably related to the higher degree of dissociation of TFSI salts, and therefore to the presence of a larger number of mobile species that are able to participate in ionic conduction. The ionic conductivity of the polymers

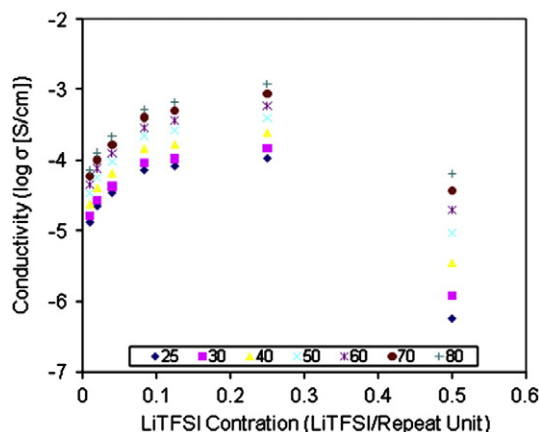
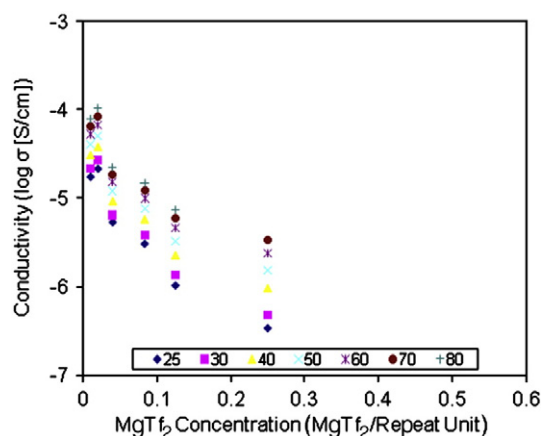


Fig. 5. Isothermal ionic conductivity of LiTFSI:MEEP.

Fig. 6. Isothermal ionic conductivity of MgTf_2 :MEEP.

with MgTFSI_2 reached its maximum at a salt molar fraction of 0.125. On the other hand, systems with LiTFSI reached a maximum conductivity at a salt molar fraction of 0.25; in other words, at twice the concentration of the electrolytes that contained MgTFSI_2 . This indicates that MgTFSI_2 has twice the charge carrying capability compared to the LiTFSI in MEEP. Classically, the cations were thought to be the dominant charge carriers due to the limitations imposed by the mass of the anion. The Mg^{2+} cation can carry twice the charge of Li^+ . Therefore the maximum conductivity of MEEP: MgTFSI_2 should be achieved at half the concentration of MEEP:LiTf. However, recent studies have suggested that the dominant charge carriers are the anion species [24]. In that case, one MgTFSI_2 molecule can dissociate to yield 2 TFSI^- anions whereas one LiTFSI molecule will yield only one. Consequently, the maximum conductivity of MgTFSI_2 will also be achieved at half the concentration of LiTFSI. The identity of the charge carrier is discussed below.

An equation that is used to describe the temperature dependence of ionic conductivity is the Vogel–Tamman–Fulcher (VTF) equation [25].

$$\sigma(T) = \sigma_0 \exp \left[\frac{-B}{(T - T_0)} \right]. \quad (1)$$

In this term, σ_0 is a pre-exponential factor related to the number of carriers. The B term is related to the E_A term in the Arrhenius equation and has the dimensions of Kelvin units, but cannot be regarded as a simple activation term. T_0 is the ideal glass transition temperature at which ion mobility goes to zero. Above T_0 , thermal motion contributes to the transport of ions and it is sometimes set as T_g but, as will be discussed later, T_0 is sometimes found to be 30–50 K below T_g .

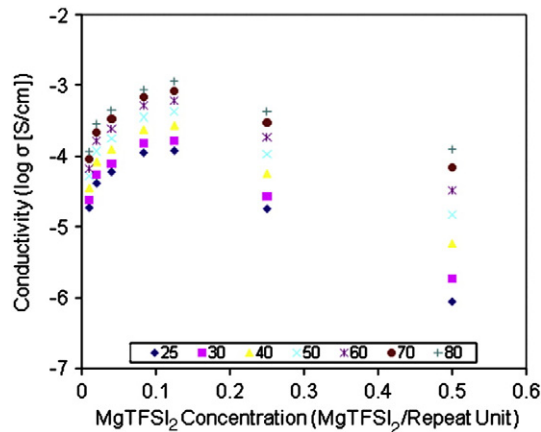
Fig. 7. Isothermal ionic conductivity of MgTFSI_2 :MEEP.

Table 1
Highest room temperature conductivity of different salts.

Salt	Highest conductivity (10^{-4} S/cm)	Salt molar fraction (salt/repeat unit)
LiTf	0.28	0.125
LiTFSI	1.06	0.250
MgTf ₂	0.22	0.020
MgTFSI ₂	1.21	0.125

During curve fitting operations, the α_0 and B parameters were set as variables. T_0 was initially set as T_g but the data points did not fit well. When T_0 was set at $T_g - 30$ K, the data points consistently fitted better. Setting T_0 to be 30–50 K below the T_g is a common practice during curve fitting of the VTF equation and has been reported previously for polyphosphazene solid polymer electrolytes [26]. If the T_0 value was set as T_g in the VTF equation, it might be assumed that the segmental motion of the polymer is the rate limiting step in ionic conduction. However, ion–ion and ion–polymer interactions cannot be neglected and both need to be overcome in ionic conduction [27]. Hence, the T_0 is set below T_g to account for the additional energy required for the transport of ions. The B parameters in the VTF equation of the SPEs were obtained and displayed in Fig. 8.

Fig. 8 shows that polymer electrolytes with the same anion show similar B values and that counterparts with TFSI salts have almost twice the B values of electrolytes with Tf salts. The strong dependence of the B value on the anionic species present suggests that the dominant charge carriers in MEEP are the anions. Assuming a minimal interaction between anion and the polymer, the B value can be described according to the Doolittle–Cohen free volume theory

$$B = \frac{f^*}{\alpha_f} \quad (2)$$

where B is the pseudo-activation energy in the Arrhenius (high-temperature) limit of the VTF relation and f^* is the averaged free volume required for transport of a molecule and α_f is the thermal expansion coefficient of the free volume [28]. The α_f value is the same for all samples because all the electrolytes contained the same polymer. Thus, the B value is only dependent on f^* . The higher B value of systems with TFSI is due to the larger volume of the TFSI[−] anion compared to the Tf[−] anion. This result is supported by observations made in previous studies where the anion was immobilized on the side chain of the polyphosphazene [29]. The conductivities of polyphosphazenes with covalently immobilized sulfonamide side groups were 10^{-6} – 10^{-7} S/cm; which is 1–2 orders of magnitude less than polyphosphazenes with free salts. When the cations were immobilized to the side chains of polyphosphazenes, the conductivities were still within the range of polyphosphazenes with free salts [30].

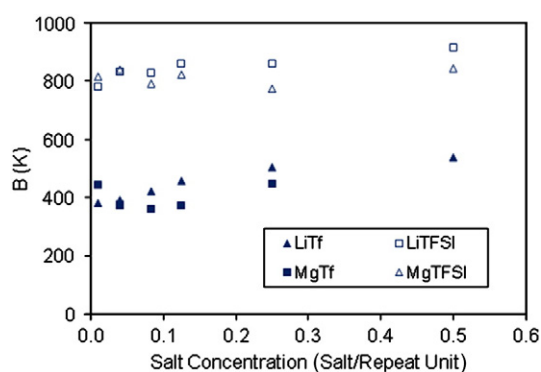


Fig. 8. VTF equation B value for MEEP polymer electrolytes.

Although the B values indicate that anions are the main contributor to ion conduction in the MEEP system, the cations are not totally immobile. Different techniques such as pulsed field gradient NMR, DC polarization, and AC impedance have shown that cations contribute to ion conduction although the transference number for cationic species is <0.5 [31]. Although polymer electrolytes prepared with magnesium salts had higher T_g s than those with lithium salts, the maximum room temperature conductivity of lithium and magnesium salts were not significantly different because the anions are the principal conductors. If the cations are strongly solvated by the polymer side groups in SPEs, any cation transport would require breaking the coordination between cations and etheric oxygens. This impedes cation mobility, and the transport of cations would depend not only on polymer flexibility, but also lability of the cation–etheric oxygen interaction [32].

5. Conclusions

The use of four different salts to study the effect of the cations and anions on ionic conduction in MEEP suggests that anions are the primary charge carriers. The high degree of dissociation of TFSI salts yielded MEEP electrolytes with higher ionic conductivities although the glass transition temperatures were also higher. Strong interactions between the cations and the polymer appear to limit the cation contribution to the overall ion conduction. However, the possibility that the two different cations interact with the polymer in different ways remains to be studied. These observations add to the information about the ion conduction mechanism in MEEP, a process that is only slowly being understood. Although this study does not support the use of magnesium salts in rechargeable battery applications, further synthetic modifications to the polyphosphazene component may enhance the mobility of Mg^{2+} ions.

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